

CPU, registers, buses and the fetch cycle ★ Golden

One college period · 40–45 minutes · Inside System Unit

What you will learn

- Describe ALU, CU, registers and cache
- Explain types of registers used in board answers
- Differentiate address, data and control buses
- Draw and explain the fetch–decode–execute cycle

Microprocessor / CPU. The integrated circuit that fetches instructions from memory, decodes them and executes them. It is the brain of the computer.

Part	Role
ALU (Arithmetic and Logic Unit)	Add, subtract, AND, OR, NOT, compare
CU (Control Unit)	Directs the flow of data; sends control signals
Registers	Very fast storage inside the CPU for one instruction/data word
Cache	Small SRAM holding recently used instructions/data
Clock	Pulses that time every step (GHz)

Registers (board favourite)

- ACC (Accumulator): stores ALU results.
- PC (Program Counter): address of the next instruction.
- IR (Instruction Register): the instruction currently being decoded.
- MAR (Memory Address Register): address being accessed in RAM.
- MDR (Memory Data Register): data coming from or going to RAM.
- General purpose registers: AX, BX, CX, DX in old x86 teaching notes.
- Flag register: zero, carry, sign bits after ALU operations.

Buses

Bus	Carries	Direction
Address bus	Memory / I/O addresses	Unidirectional (CPU → memory)
Data bus	Actual data and instructions	Bidirectional
Control bus	Read, write, interrupt, clock	Mostly CPU → others (some lines back)



repeats for the next instruction

Instruction cycle. Board wording: Fetch Cycle (often they mean the whole fetch-execute cycle).

LEARN IT

Fetch: PC gives address, instruction copied from memory into IR, PC incremented. Decode: CU interprets the opcode. Execute: ALU or memory operation happens. Store/write-back: result goes to a register or RAM. Then the cycle repeats.

HOW BIEK ASKS THIS

2026 Q2(iv): "What is fetch Cycle? Draw its diagram." Draw four boxes Fetch→Decode→Execute→Store with a loop arrow. Q3 OR: Computer Register and its types.

REMEMBER IT

Clock speed, word size (32/64 bit), cache size and number of cores decide CPU power. Generation: 1st vacuum tubes ... 3rd ICs ... 4th microprocessors ... 5th AI.

اردو میں سمجھیں

CPU میں ALU حساب کرتی ہے، CU حکم دیتی ہے، رجسٹر بہت تیز چھوٹی میموری ہیں۔
Fetch سائیکل: میموری سے ہدایت لاؤ، سمجھو، چلاؤ، نتیجہ رکھو۔ ایڈریس بس پتہ لے جاتی ہے، ڈیٹا بس ڈیٹا، کنٹرول بس اشارے۔

CHECK YOURSELF

1. Which register holds the next instruction address?
2. Is the data bus uni- or bi-directional?

Answers: 1. Program Counter (PC) · 2. Bidirectional

MCQs

1. The third generation of computers used:
 - a. Vacuum tubes
 - b. Transistors
 - c. Integrated circuits
 - d. Microprocessors
2. ALU is a part of the:
 - a. Printer
 - b. CPU
 - c. Monitor

d. Keyboard

Key: 1=c, 2=b

Short questions (3 marks each)

1. What is the fetch cycle? Draw its diagram.
2. What is a computer register? Explain its types.
3. Differentiate address bus and data bus.

Model short answers

1. The CPU repeatedly fetches an instruction from memory, decodes it, executes it and stores the result. Diagram: Fetch → Decode → Execute → Store, looping.
2. A register is a small high-speed memory cell in the CPU. Types: accumulator, PC, IR, MAR, MDR, flags, general purpose.
3. Address bus sends locations one way from CPU; data bus carries the bits both ways.

Long questions (board Section C)

1. Explain the internal architecture of a microprocessor. Draw and describe the fetch-execute cycle.